

Lab W3D2

Question 1. Starting with the values 1, 3, 4, 4, 5, 7, 9, 11, 13, 13, 17, do the following:

- a. Create a heap H in which these values are the keys by inserting one item at a time. You must show the heap after each insert both as a tree as well as array.
- b. Create a heap H in which these values are the keys by the bottom up approach. You must show the heap after each insert each new value is placed in the heap in both as a tree and as an array.

Question 2. Carry out the array-based version of HeapSort on the input array [1, 5, 2, 11, 12, 2, 3]. **You need to first build the heap bottom up. You need not show all the steps.**

Question 3. Given two Heaps H1 and H2 of the same height, give an efficient algorithm to “fuse together” to make one Heap.

- (a) What is the best case?
- (b) What is the worst case?
- (c) What is the worst case time complexity?